Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	*Aretz, Martin*	Email	*martin.aretz@ipt.fraunhofer.de*
Institute Name	*Fraunhofer Institute for Production Technology IPT*			
Address	*Steinbachstrasse 17, 52074 Aachen, Germany*			

2. Technology Overview

Title of Proposed Technology	*Optimized production of expanded metal based PTLs for PEM-electrolyzers*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery / materials		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Cutting Machine Tools Other Precision Manufacturing Machinery–related Technologie*	*New & Renewable Energy*	*Hydrogen*
Concept of Proposed Technology	*Development, production and testing of a cost- and material-optimized manufacturing process for multi-layer porous transport layers for electrolyzers, providing improved fluidic, mechanical and electrical performance by targeted modification of process parameters and component design.*			
Rationale for Joint R&D Project	*A joint R&D project is essential, as Korea has identified hydrogen as a strategic pillar for decarbonizing its industry, mobility and power sectors and is investing heavily in large-scale electrolyzer deployment. Expanded metal is a particularly promising technology for electrolyzers that, when properly designed and manufactured, can significantly lower hydrogen production costs by reducing material use, improving efficiency and enabling scalable, automated production. Korean partners bring strong capabilities in advanced materials, high-volume manufacturing and*			

	industrialization, while Fraunhofer IPT contribute complementary strengths in process development and component design. Working together will accelerate the development and validation of cost-effective porous transport layers, creating benefits for both the Korean and European hydrogen economies.
Key Words	1. *Production of stack components of electrolyzer* 2. *Reduction of hydrogen production cost through CAPEX reduction* 3. *Designing and resistance welding of expanded metal structures* 4. *Ex-Situ and In-Situ characterization of innovative PTL structures* 5. *Porous Transport Layer (PTL)*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (6)
R&D Objectives	*Development, production and testing of a cost and material optimized production process for multi-layer porous transport layers for electrolyzers with optimized fluidical, mechanical and electrical characteristics by modification of the production process and component design.*			
R&D Target Specifications	1. *Characterization of an electrode structure (extended metal - bipolarplate - extended metal) for electrolyzers, that is optimized for production* 2. *Process development for single layer expanded metal development* 3. *Process development for welding of extended metals e.g. geometry as well as the order of the base material, electrode geometry and process parameters to eliminate the need for coating* 4. *Investigation of the Interaction between extended metal geometry and welding process* 5. *Evaluation of single cell performance (U-I-curve) depended of the component design and production process*			
R&D Plan	*Project time 3.0 years / 12 Quarters; Q= Quarters needed for Workpackage* ***WP1 Requirements analysis and definition of relevant structures (Q1-2)*** *1.1 Derive KPIs from goal and target specs (fluidic, mechanical, electrical, cost, material use).* *1.2 Define design space for "expanded metal – bipolar plate – expanded metal" structures.* *1.3 Specify test methods and reference setups (mechanical, electrical, fluidic, single-cell).* ***WP2 Design and fabrication of production-optimized electrode structures (Q2-5)*** *2.1 Pre-design geometries using simple FEM/CFD and manufacturability constraints.* *2.2 Fabricate first prototypes with existing processes and materials.* *2.3 Characterize structures (ex-situ) and derive design rules for production-optimized layouts.* ***WP3 Process development for single-layer expanded metal (Q2-6)*** *3.1 Define target window for geometry and properties vs. cost/material usage.* *3.2 Systematically vary expansion and rolling parameters, material thickness and grade.* *3.3 Deliver parameter sets and control strategy for reproducible single-layer production.* ***WP4 Welding process and geometry–process interaction (Q3-8)*** *4.1 Screen resistance welding technologies and select candidates for thin expanded metals.* *4.2 Design multi-layer concepts (graded porosity, support + distribution layers).* *4.2 Analysis of influence of mesh geometry, stack order, electrode shape and process parameters.* ***WP5 Characterization of multi-layer PTLs and electrochemical testing (Q9-12)*** *5.1 Test in single cells (U–I curves, pressure drop, durability) vs. references.* *5.2 Correlate design/process parameters with fluidic, mechanical and electrical performance.* ***WP6 Partner technology benchmarking and scale-up roadmap (Q9-12)*** *6.1 Partner benchmarks own PTLs vs. new ones (performance, cost, material use, yield).* *6.2 Identify retrofit options for existing lines and product platforms.* *6.3 Define industrialization roadmap (CAPEX, timeline, risks, quality standards).*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. *Support in the development of a cost effective production process for PTLs in electrolyzers* 2. *Execution of expanded metal and resistance welding process (own production inhouse)* 3. *Knowledge transfer in the production of electrolyzer components e.g. bipolar plates and PTLs* 4. *Ex-situ (functional) characterization of innovative electrode structure* 5. *In-situ single cell test for evaluation of electrochemical performance of electrode structure*
Expected R&R of Korean SME Partner	1. *Investigation of the impact of process development on upstream or downstream processes* 2. *Benchmark expanded metal solutions with own PTL manufacturing technology* 3. *Input of further system requirements for the production process* 4. *Utilization of process data to improve quality assurance*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. *Optimization of existing production processes for multi-layer porous transport layers and other electrolyzer components* 2. *Significant reduction of production costs and material usage through tailored expanded metal design and welding processes* 3. *Validation of high-performance, scalable PTL concepts in single-cell tests for industrial electrolyzer applications* 4. *Enabling access to new markets and joint commercialization opportunities in the global hydrogen and electrolyzer sector*